The conformal point is a phase boundary: unfolding directions, the sign map of the cascade divisor, and a two-channel photonic readout

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Abstract

The equal-frequency Pais–Uhlenbeck point—the per-mode kinematics of conformal gravity—is an exceptional point (EP), and this paper treats it as what an exceptional point is: a phase boundary. We classify every deformation of the theory by two data: the *kind* of unfolding direction (family-parameter detunings split the degenerate pair linearly; generic matrix perturbations split by Puiseux $\sqrt{\epsilon}$) and its *phase*, which decides oscillatory versus growing dynamics. On this map the oscillatory set is the punctured real axis ($\Delta \neq 0$) of the family directions (while ω_2^2 stays positive) and, the ray $\varphi = 0$ (the single-entry-model reference direction, resolved to within $O(\epsilon)$) of the generic ones—measure zero (in the complex plane) either way (the family real axis by the analytic law; the generic ray by the exact Puiseux eigenvalues, from which growth with probability one follows for any absolutely continuous phase law (unstructured, traceless/phase-only regulators; fully generic noise still gives probability $\to 1$, at rate $1 - O(\sqrt{\epsilon})$ for phase densities bounded near the band)—corroborated by random-phase perturbations landing on the growing side in 500/500 draws at $\epsilon = 10^{-8}$), so any *unstructured* regulator of the pure theory buys the growing side. Real-symplectic (structure-preserving) regulators are the sole exception—the characteristic polynomial stays even and real, so its discriminant sign splits perturbation directions into two *open* half-spaces, an oscillatory band and a growing one, and which side is bought then has probability strictly between 0 and 1 (developed below). The known deformations take their places: an Einstein term unfolds on the real axis (Stelle, oscillatory, dictionary cost $2/\delta$); the Donoghue–Menezes loop width tilts the approach ray by a fixed angle without imposing a detuning floor on the approach to the exceptional line; and the Ilhan–Kovner "ghost-turned tachyon" blowup of conformal gravity's $k = 0$ sector is the corner case $\omega \to 0$, where the family direction itself turns Puiseux—their finite-time singularity sits on the corner's imaginary-split unfolding (the $\varphi = \pi$ direction, where the detuned $\omega_2^2 < 0$); their blowup is reproduced here with no free parameters, while the map's frozen-mode rate $(\sqrt{3}/2)\gamma_b$ is the rate their equations imply once their effective detuning is ingested. The boundary is then shown to be readable at the level of the theory's stochastic (beable) representation: for cascade (triangular) processes we prove the divisor of the process is triangular with a per-bond sign law; that within Jordan cascades the pure conditional-mass-creation mechanism of the companion paper is strictly a property of the minimal block—for Jordan blocks of size $k \geq 3$ the divisor generator has entries of *both* signs, in an exactly alternating pattern given by the Rayleigh sums of the zeros of the Bessel function J_0. At finite steps a dominant-pole argument—asymptotic, with numerically computed window edges, below the strength of the theorems—locates the divisor's negative windows on Fourier–Bessel bands with edges at $(j_{0,n}/j_{0,1})^2$. Real detuning brings negativity back at the per-bond first-return length $\pi/|\delta_i|$; for splits in the closed upper half-plane the recurrence is paced by the *real part* of the split, while the negativity-free directions comprise the +i ray, the EP, and a computed

silent cone centered on the $-$i ray whose sharp thresholds reach to within $\approx$15.3° of the real axis, from below (silent split phases $\approx$195.3°–344.7°; thresholds located to 10^{-3} degrees, cone interior sampled, in the text): the negativity channel is a spectrometer for the magnitude of the real part of the split, with a known silent sector. The divisor's second diagnostic—the column-sum (conditional-mass-creation) channel, stated relative to the minimal contraction normalization defined in the text—fires at every sampled direction (a 5° scan of the full circle finds no silent phase) and its onset, at known $|\delta|$, resolves the growing from the decaying imaginary ray (relative growth of the fed level, under the symmetric split convention fixed in the text)—a sign the negativity channel cannot see; and the $\omega \to 0$ corner, being a size-4 Jordan block rather than a bond, is negativity-bearing by the $k \geq 3$ theorem at the corner itself and, computationally, at every sampled unfolding direction. The two channels together leave no sampled direction silent, bond by bond. A two-device photonic extension of our measurement proposal (a coupler *at* its EP, and a dissipatively-coupled unidirectional pair) would measure the mechanism fingerprint and the direction spectroscopy on capabilities demonstrated in the platform family, one coupling transfer priced into the device error budget. All results are tree-level mode kinematics, stated for the minimal cell and its finite stackings.

Note on provenance. This manuscript is an experiment in AI-conducted research as much as a physics paper. The project began when the author asked the large language model Claude Fable 5 (Anthropic): *"Is it not possible that the antilinear PT/time-reversal symmetry element could maybe predict indivisible non-Markovian processes?"* Everything that followed was performed by the model: the research program, the theorems and proofs, the numerical experiments, the literature search, and this text. The model was repeatedly prompted to attack its own results adversarially and to search for prior work. The author, a professional engineer rather than a physicist, directed and reviewed the sessions but makes no claim to comprehension of the findings, and is circulating the manuscript for independent review by professional physicists. Accordingly, nothing here should be taken on trust: every theorem is stated with a full proof, and every numerical claim is reproducible from the public scripts, with one disclosed exception (the Merlin row of Sec. 4, which is analytic—its arithmetic stated in the revised companion; see Code availability); the originating conversation, project briefing, and full research notebook are preserved in the program's hub repository, `https://github.com/embeddetech/ghosts-program`.

1 Introduction

This is the sixth paper of a program that reads quantum dynamics in Barandes' currency of indivisible stochastic processes on configurations [13] and asks what becomes of ghost sectors there [1, 2, 3, 4]. The conformal-gravity paper [4] established the arena for the present one: the equal-frequency Pais–Uhlenbeck (PU) mode pair—the kinematics that conformal gravity carries at every spatial momentum—is a Jordan block, an exceptional point (EP) of the mode dynamics, and at that point no closed stochastic representation of any kind exists, while the open, survival-conditioned representation is a one-way cascade whose indivisibility enters through a new mechanism (*conditional mass creation*: the two-point divisor stays entrywise nonnegative but its column sum crosses one, a diagnostic stated relative to the minimal contraction normalization made explicit in Sec. 3) [4].

That paper treated the point itself. This paper treats its neighborhood, and the organizing claim is the title's: *the conformal point is a phase boundary*, and everything in its vicinity is classified by how a deformation unfolds the Jordan block. The claim has three layers.

Layer 1: the direction map (Sec. 2). Deformations come in two kinds. Physical parameter changes—an Einstein–Hilbert term, a loop-induced width—move within the PU *family*, detuning $\omega_2^2 = \omega^2 + \Delta$ and splitting the degenerate frequencies *linearly* in $|\Delta|$ (the non-generic unfolding responsible for the $2/\delta$ dictionary cost computed in [4]). Generic *unstructured* perturbations—floating-point noise and random regulators that touch the matrix rather than the family (a structured real-symplectic lattice is the exception treated below, with an open oscillatory band)—split by Puiseux $\sqrt{\epsilon}$ [8]. In both kinds the *phase* of the perturbation decides the dynamics, with a kind-dependent oscillatory set (within an $O(\epsilon)$ neighborhood of the positive-real ray for traceless perturbations; the punctured real axis ($\Delta \neq 0$), while $\omega_2^2 > 0$, for the family) and growth everywhere else (for net-damping fully generic noise an $O(\sqrt{\epsilon})$-wide decaying band intervenes, so growth there holds only with probability $1 - O(\sqrt{\epsilon})$). The oscillatory set is measure zero in the generic directions, and the $\omega \to 0$ corner—conformal gravity's homogeneous sector—is where even the family direction turns Puiseux: the Ilhan–Kovner instability [6] is recovered there, quantitatively, as the corner's imaginary-split unfolding.

Layer 2: the divisor image (Sec. 3). The program's currency sees the same boundary. For cascade processes we prove a per-bond sign law for the two-point divisor and show that the sign structure of a deep Jordan cascade's divisor is universal Bessel-zero arithmetic: the divisor

generator's entries alternate in sign with coefficients given exactly by the Rayleigh sums $R_p = \sum_n j_{0,n}^{-2p}$, and finite-step negative windows converge to Fourier–Bessel bands (a dominant-pole asymptotic, below theorem strength). Two corollaries matter physically: the mass-creation-pure mechanism of [4] is confined to the minimal ($k = 2$) Jordan cell—no taller Jordan cascade is negativity-free—and a real-detuned cascade's divisor develops negativity at the first-return length $\pi/|\delta_i|$ per bond, the recurrence for upper-half-plane splits being paced by the real part of the split (a cone about the $-$i ray is silent for negativity; Sec. 3). The negativity channel alone is therefore a spectrometer for the magnitude of the *real* part of the split; its silent set—the $+$i ray, the EP, and the cone about the $-$i ray—is read out at every sampled phase by the divisor's second, convention-relative channel (the column sum, silent at no sampled phase), whose onset resolves the sign negativity cannot see, and the $\omega \to 0$ corner—a size-4 block—is negativity-bearing by the $k \geq 3$ theorem before any unfolding is chosen (Sec. 3): a two-channel direction readout.

Layer 3: the readout (Sec. 5). Both the mechanism fingerprint and the direction spectroscopy are measurable with classical light on the femtosecond-laser-written waveguide platform of Klauck et al. [17], via a two-device extension of our standing proposal: a coupler operated exactly at its EP, and a dissipatively-coupled pair at the unidirectional point that realizes the PU cascade cell in the guide basis.

Throughout, the honest scope is the one inherited from [4]: these are tree-level statements about mode kinematics, proved for the minimal cell and finite stackings, with the interacting field theory entering only through the deformation dictionary (Sec. 4).

2 The direction map

Write the mode pair in its normal form. At the conformal point the one-particle matrix of a mode is similar to the 2×2 Jordan cell $J = \omega\mathbf{1} + N$, $N_{12} = 1$. A deformation acts either *within the PU family*—the physical parameters (ω_1^2, ω_2^2) move, $\omega_2^2 = \omega^2 + \Delta$ with $\Delta = |\Delta| e^{i\varphi}$—or *generically*, as a matrix perturbation, modeled by the block entry $H = \left(\begin{smallmatrix} \omega & 1 \\ \epsilon e^{i\varphi} & \omega \end{smallmatrix}\right)$.

Family direction: linear split. The eigenfrequencies of the detuned pair are ω and $\sqrt{\omega^2 + \Delta} \simeq \omega + \Delta/2\omega$: the split is linear in $|\Delta|$ (the family direction is a non-generic unfolding; this is why the dictionary cost of [4] scales as $2/\delta$ rather than $2/\sqrt{\delta}$, re-verified below). The growth rate is

$$\lambda_{\mathrm{fam}}(\Delta) = \left|\mathrm{Im}\sqrt{\omega^2 + \Delta}\right| \;\simeq\; \frac{|\Delta|\,|\sin\varphi|}{2\omega}, \tag{1}$$

verified on the 4×4 PU companion matrix at $|\Delta| = 10^{-3}, 10^{-2}, 10^{-1}$, $\varphi \in \{0, \pi/6, \pi/2, 5\pi/6, \pi\}$—the lower half-plane follows by conjugation symmetry, $\lambda(\bar{\Delta}) = \lambda(\Delta)$ (worst relative deviation 4.6×10^{-2} over the nonzero-growth phases, dominated by the $|\Delta| = 10^{-1}$ curvature; Eq. (1) is the $|\Delta| \to 0$ law). Two features deserve emphasis. First, the oscillatory set of the family direction is the punctured *real axis* of Δ ($\Delta \neq 0$, both signs), so long as $\omega_2^2 > 0$: at $\varphi = 0$ and $\varphi = \pi$ the computed growth is $\lesssim 10^{-13}$. Second, on the real axis the EP approach carries the known cost: the eigenvector-matrix condition number obeys $\mathrm{cond}(S)\,\delta/2 = 1.008$ and 1.001 at $\delta = 5 \times 10^{-3}, 5 \times 10^{-4}$—the $2.00/\delta$ law of [4], recovered here as the boundary-line toll.

Generic direction: Puiseux split. The perturbed block has eigenvalues $\omega \pm \sqrt{\epsilon}\, e^{i\varphi/2}$, so

$$\lambda_{\mathrm{gen}}(\epsilon) = \sqrt{\epsilon}\,\left|\sin(\varphi/2)\right|, \tag{2}$$

verified to 1.2×10^{-13} across the grid, with the fitted Puiseux exponent 0.5000. Here the oscillatory set is the single ray $\varphi = 0$ of this single-entry model (a fully generic matrix perturbation adds a common $O(\epsilon)$ growth $\frac{1}{2}\epsilon \operatorname{Im}\operatorname{tr} P$, absent here where $\operatorname{tr} P = 0$; the oscillatory locus is instead displaced from this ray by only $O(\epsilon)$ through the sub-leading discriminant correction (and is empty once $\operatorname{Im}\operatorname{tr} P \neq 0$, its role taken by the growing/decaying split below) and, being contained in the zero set of a single real-analytic condition (the imaginary part of the discriminant, not identically vanishing in the phase), is measure zero a fortiori): a random-phase perturbation at $\epsilon = 10^{-8}$ landed on the growing side in 500 of 500 draws, with median rate $\sim \sqrt{\epsilon}$. *The pure theory is the boundary* is thus an analytic statement—growth with probability one for any absolutely continuous phase law, from the exact Puiseux eigenvalues—corroborated by sampling: an *unstructured* regulator that perturbs the block with a continuously-distributed phase (floating-point roundoff, modeled as an effectively random phase over inputs, being the representative case) buys the growing side with probability one, at the Puiseux-enhanced rate $\sqrt{\epsilon}$; a fully generic (non-traceless) perturbation instead opens, when the common term damps ($\operatorname{Im}\operatorname{tr} P < 0$), a decaying band of angular width $O(\sqrt{\epsilon})$ through the common $\frac{1}{2}\epsilon \operatorname{Im}\operatorname{tr} P$ term, so there the growing side has probability $\to 1$ as $\epsilon \to 0$ (at rate $1 - O(\sqrt{\epsilon})$ for phase densities bounded near the band). Structured regulators are a different matter: a real-symplectic lattice discretization sits at a defective 1:1 resonance, whose Hamiltonian unfoldings keep an open set of directions oscillatory (a property of the real-symplectic unfolding, not of a Krein signature—undefined at the defective point [9]), so there, for any direction law with full support, the growing side has probability strictly between zero and one (an open band of oscillatory directions survives); and a deterministic truncation lands at one definite phase, for which "probability one" is not the right description.

The $\omega \to 0$ corner. Conformal gravity's translationally-invariant ($k = 0$) sector sits at the doubly degenerate corner $\omega_1 = \omega_2 = 0$, with secular solution set $\{1, t, t^2, t^3\}$. There the family direction itself becomes Puiseux: the induced frequencies are $\pm\sqrt{\Delta}$, so $\lambda_{\text{corner}} = \sqrt{|\Delta|}\,|\sin(\varphi/2)|$ (verified to 6.9×10^{-16}). The two phases in play are related by the square root: on the finite-frequency line the split is $\delta \simeq \Delta/2\omega$ and shares the phase of Δ, while at the corner $\delta = \sqrt{\Delta}$ *halves* it—which is how a $\varphi = \pi$ family deformation, real in Δ, arrives as a purely imaginary frequency split (the +i ray of Sec. 3's split-phase classification, on the principal branch; the corner block itself sits outside the bond-level classification, as Sec. 3 shows). This corner is where the Ilhan–Kovner result lives. In [6] the nonlinear self-interaction of conformal gravity's $k = 0$ sector generates, for a background amplitude γ_b, an effective potential whose sign relative to the ghost kinetic term is anti-restoring: in our variables, $\Delta_{\text{eff}} = -\frac{3}{4}\gamma_b^2$, the $\varphi = \pi$ ray. With their Δ_{eff} ingested, the map returns growth at $\sqrt{3/4}\,\gamma_b \cdot |\sin(\pi/2)| = (\sqrt{3}/2)\gamma_b$—algebra once Δ_{eff} is given, not an independent prediction—and the frozen-mode eigenvalue check confirms rates 0.433013/0.866025/1.732051 at $\gamma_b = 0.5/1/2$, matching $(\sqrt{3}/2)\gamma_b$ to the six digits shown. The parameter-free corroboration is separate: direct integration of their *nonlinear* equations of motion (public script, block [3b]) reproduces their exact solution, Eq. (58) of [6], to relative 5×10^{-12} and the singularity time $t_{\text{sing}} = 2^{3/2}/\gamma_0$ (γ_0 their initial amplitude) to relative $\sim 10^{-6}$ (five–six significant digits). Their "ghost-turned tachyon" is the imaginary-split unfolding of the conformal corner (a $\varphi = \pi$ family direction whose frequency split is imaginary because $\omega_2^2 < 0$ there); the reproduction, not the rate, is what carries "no free parameters."

Three scope guards, which we owe to a careful reading of [6, 7]. Their exceptional point is the $\omega = 0$ corner, not the finite-frequency line: the extension of the classification to finite k is ours, not theirs. Their detuning is *self-induced* and amplitude-dependent ($\Delta_{\text{eff}} \propto \gamma_b^2$), so the unfolding statement is about the interacting background, never about the linear spectrum. And

the companion paper [7] exhibits an interacting particle–ghost system that remains stable—in our language, an interaction that unfolds along the real direction—so "interactions destabilize ghosts" is not a universal; *which way the interaction unfolds the block* is the invariant content, and their stable example is a test case the classification passes rather than an exception it suffers.

3 The divisor image

The companion papers measure dynamics in the beable currency: a mode-pair evolution $U(t)$ is read through $\Gamma(t) = |U(t)/\sigma_{\max}(t)|^2$ (entrywise moduli of the contraction-normalized evolution), and divisibility is interrogated through the two-point divisor $M = \Gamma(s+\tau)\,\Gamma(s)^{-1}$. The indivisible-process paper [1] found the broken dimer fails sub-stochasticity by an entry going *negative*; the conformal-gravity paper [4] found the PU cell fails by the column sum crossing one with nonnegativity intact (*conditional mass creation*). This section proves where each mechanism lives.

Conventions. Three conventions are inherited from [1, 4] and restated here, because the results below split cleanly by whether they depend on them. The *beable basis* is the configuration basis in which the cascade Hamiltonian is triangular; processes are read entrywise, $\Gamma = |\cdot|^2$; and $\sigma_{\max}(t)$ denotes the largest singular value of $U(t)$, so that $C(t) = U(t)/\sigma_{\max}(t)$ is the *minimal contraction normalization*: a unitary (conservative) dilation exists exactly when the normalized evolution is a contraction, so $\sigma_{\max}$ is the pointwise-least normalization admitting a conservative completion—the unique choice that discards no survival probability beyond what the dynamics itself loses ($\|C(t)\| = 1$: at every time the least-decaying state is not conditioned at all). The two failure diagnostics then differ in kind. Divisor entry *signs* are invariant under every positive scalar renormalization $c(t)$ of the normalization—negativity (M1) is convention-free, as Proposition 1(ii) records—while divisor *column sums* scale by $c^2(s)/c^2(s+\tau)$: the column-sum-crossing diagnostic (conditional mass creation, M2) is stated relative to the $\sigma_{\max}$ convention, and its onset—indeed its occurrence—moves under renormalization. The relativity is quantifiable, and quantifying it is what justifies the convention: for the unit cell with coupling κ the raw $|U|^2$ divisor's column sum over the step from s to $T = s+\tau$ is exactly $1 + \kappa^2(T^2 - s^2)$, so crossing-freedom at the pair (s, T) forces $c^2(T)/c^2(s) \geq 1 + \kappa^2(T^2 - s^2)$; telescoping this over a refined partition of $[0, t]$ turns the product into the Gaussian, so a renormalization avoids the crossing at *all* step pairs only if $c^2(t) \geq c^2(0)\,\mathrm{e}^{\kappa^2 t^2}$, and that Gaussian envelope does avoid it (the exact column-sum law and the envelope's sufficiency verified in `scripts/p6_divisor_gate.py`, block [4]; the necessity is the telescoping just given). Divisibility can thus be purchased—within positive scalar renormalizations $c(t)$—but only by conditioning away survival probability at a Gaussian rate against polynomially growing dynamics—a conditioning overhead $c^2/\sigma_{\max}^2$ that diverges; under the minimal convention, in which conditioning means survival and nothing else, the crossing is unavoidable. Convention-free content below: the triangularity of the process and every divisor sign statement (Proposition 1, Theorems 1 and 2). Convention-relative: every mass-creation onset (t_c, z_c).

Notation. From here on, δ (with per-bond subscripts δ_i, and as the device detuning of Sec. 5) always denotes a *frequency* split, as in Sec. 2; the symbol Δ is reserved for Sec. 2's family detuning, which lives in ω^2 ($\omega_2^2 = \omega^2 + \Delta$). On the finite-frequency line $\delta \simeq \Delta/2\omega$; at the corner $\delta = \sqrt{\Delta}$ halves the phase.

Proposition 1 (cascade divisor; per-bond sign law)**.** *Let H be upper triangular on d configurations (a cascade), and let $V(t) = |U(t)|^2$ entrywise, $U(t) = \mathrm{e}^{-\mathrm{i}Ht}$. Then for all $0 < s < s+\tau$:*

(i) the divisor $M = V(s+\tau)V(s)^{-1}$ is upper triangular with strictly positive diagonal $M_{ii} = V_{ii}(s+\tau)/V_{ii}(s)$; and (ii) for each bond i,

$$\operatorname{sign} M_{i,i+1} = \operatorname{sign}\big[r_i(s+\tau) - r_i(s)\big], \qquad r_i(t) = \frac{|U_{i,i+1}(t)|^2}{|U_{ii}(t)|^2},$$

independently of the $\sigma_{\max}$ normalization.

Proof. $U(t)$ is upper triangular, hence so are V, V^{-1}, and M; the diagonal of M is the ratio of positive diagonals. For (ii), the inverse of an upper-triangular matrix has $(V^{-1})_{i,i+1} = -V_{i,i+1}/(V_{ii}V_{i+1,i+1})$, so

$$M_{i,i+1} = \frac{V_{i,i+1}(s+\tau)}{V_{i+1,i+1}(s)} - \frac{V_{ii}(s+\tau)\,V_{i,i+1}(s)}{V_{ii}(s)V_{i+1,i+1}(s)} = \frac{V_{ii}(s+\tau)}{V_{i+1,i+1}(s)}\big[r_i(s+\tau) - r_i(s)\big],$$

a positive multiple of the ratio increment. The $\sigma_{\max}$ normalization multiplies M by a positive scalar. □

Adversarial check: 1408/1408 sign agreements over random complex triangular families at $d = 3, 4, 5$, with the lower-triangular residual exactly zero.

For the detuned 2×2 cascade $H = \left(\begin{smallmatrix} \omega_1 & c \\ 0 & \omega_2 \end{smallmatrix}\right)$ the ratio is explicit: $r(t) = |c|^2\big(2 - 2\cos\delta t\big)/\delta^2$ for real split $\delta = \omega_1 - \omega_2$ (oscillates: negativity *recurs*, first onset $s^* = \pi/|\delta|$; computed onset ratios 0.9969–0.9995 over $\delta = 0.4$–2.0, at a 0.02 divisor step with a 0.01 base-point grid—the detected onset is the first grid point past $\pi/|\delta| - \tau/2$); $r(t) = |c|^2\big(1 - \mathrm{e}^{(g_1-g_2)t}\big)^2/(g_1-g_2)^2$ for a decay split (monotone: negativity-free at all times, machine epsilon); and $r(t) = |c|^2 t^2$ at the EP (monotone: the mass-creation-pure cell of [4], onset re-verified at 0.48 at a 0.05 pair spacing with a 0.01 base grid; the zero-step limit is 0.486, of which the companion's ≈ 0.50 was its coarser 0.05 base grid rounding up—finite-pair conventions can shift the detected onset in either direction). The general split $\delta = a + \mathrm{i}b$ interpolates: $r(t) = |c|^2|\mathrm{e}^{\mathrm{i}\delta t} - 1|^2/|\delta|^2$, whose derivative is $\propto \mathrm{e}^{-bt}\big[b\cos at + a\sin at - b\,\mathrm{e}^{-bt}\big]$. For $b \geq 0$ (splits in the closed upper half-plane) this changes sign whenever $a \neq 0$, so there the negativity-free directions are exactly the $+\mathrm{i}$ ray and the EP, and *the negativity channel reads out the magnitude of the real part of the split*: real-direction unfolding brings negativity back with first-return time $\pi/|\delta|$, diverging as the EP is approached, and a mixed split recurs on the pace of $|\operatorname{Re}\delta|$ (its sign is a residual ambiguity of the negativity channel). For $b < 0$ the term $-b\,\mathrm{e}^{-bt} = |b|\mathrm{e}^{|b|t}$ outgrows the oscillation amplitude $\sqrt{a^2+b^2}$ at $t^* = \ln\big(\sqrt{a^2+b^2}/|b|\big)/|b|$, and when t^* precedes the first trough the ratio is, heuristically, monotone despite $a \neq 0$ (the thresholds below come from numerical bisection, not from this sufficient condition): the negativity-free set acquires a *silent cone* (interior sampled) centered on the $-\mathrm{i}$ ray, spanning split phases 195.345°–344.655° (sharp, scale-invariant thresholds within 15.345° of the real axis (from below); bisection to 10^{-3} degrees; at the thresholds themselves the derivative is tangent to zero, so they are silent). The wedges between the real axis and the thresholds are negativity-bearing but *transient*: finitely many windows before the growth term silences the bond—a single window over the few degrees adjacent to the threshold, proliferating toward the real axis, where the silencing time t^* diverges (computed, at the sampled wedge phases: 1 window at 190° and 193°, 2 at 188°—completed totals, scale-invariant—and 4 within $t \leq 40$ at $|\delta| = 0.6$ at 181° and 185°). A wedge direction is thus readable only in a finite early epoch, in contrast with the indefinitely recurrent upper half-plane. Computed over the full circle at $|\delta| = 0.6$, t on the uniform grid 0.05:0.02:40: monotone (0 decreasing intervals) at 90° and at the sampled phases 210°–330° (interior to the computed silent cone, whose connectedness we read

from this sampled grid rather than a dense scan), versus several hundred to $\sim 10^3$ decreasing intervals at the sampled phases $0°, 30°, 60°, 120°, 150°, 180°$ (919 at $30°/150°$, 953 at $0°/180°$; the near-threshold count at $60°/120°$ is tolerance-sensitive in the exponentially flattened tail, its implementation-robust strict count being 1050). The negativity channel is thus a spectrometer for the magnitude of the real part of the split, with a known silent sector. In a real-detuned bidiagonal chain the readout is per bond: each bond i returns at $\pi/|\delta_i|$ (computed ratios 0.9982–0.9995, $d = 4$).

The second channel, and the corner. The silent directions of the negativity channel are not silent for the divisor as a whole. The column-sum diagnostic—conditional mass creation, convention-relative in the sense of the conventions paragraph—fires at every *sampled* split phase: computed onsets over the full circle at $|\delta| = 0.6$ (a 5° grid, 72 phases; coupling 1, pair step 0.05, base grid 0.01; `scripts/p6_m2_channel.py`) range from 0.30 to 0.68 with no silent phase found, and on the imaginary axis the onset is *asymmetric*—0.68 on the growing (+i) ray against 0.30 on the decaying (−i) ray (0.58 vs 0.38 at $|\delta| = 0.3$; 0.69 vs 0.30 at pair step 0.02)—so the second channel, at known $|\delta|$, resolves the sign of the *imaginary* part (the ±i ray); both channels are even under $\mathrm{Re}\,\delta \to -\mathrm{Re}\,\delta$, so the sign of the *real* part stays invisible to either. With the split assigned symmetrically, $\omega_{1,2} = \omega \pm \delta/2$ (as in the scripts; the beable process is invariant under a common frequency shift, so δ alone does not fix an absolute assignment), the +i ray is the direction in which the fed (downstream) level grows relative to its source; "growing/decaying ray" is shorthand for this relative sense. And the flagship corner instability of Sec. 2 never relied on the bond-level statement in the first place: the $\omega \to 0$ sector is a size-4 Jordan block, not a bond, so Theorem 1 applies, and its divisor is negativity-bearing at depth two for every sufficiently short step—by the theorem for the corner block itself, and computationally at every sampled unfolding direction. Computed on the corner companion matrix (base grid 0.002), the negativity onset sits at the first grid point past the theorem's short-step boundary $s^* = \tau/(\sqrt{3}-1)$ (0.070 vs $s^* = 0.0683$ at $\tau = 0.05$; 0.028 vs 0.0273 at $\tau = 0.02$), identically for the pure corner, the $\varphi = \pi$ (Ilhan–Kovner) unfolding, and the $\varphi = 0$ real-split one—and identically, onset 0.070, at all twelve sampled phases of a full unfolding circle at $|\Delta| = 0.36$—while the column-sum onsets differ by direction (0.800, 0.906, 0.728 respectively at $\tau = 0.05$). The instrument is thus two-channel: negativity reads the real part of the split and is silent on the imaginary directions at the bond level; the column sum—stated in the minimal convention—fires at every sampled direction and resolves the imaginary sign; and the deep-block corner announces itself through the sign theorem before any unfolding is chosen.

Purity, however, is a property of the *minimal Jordan* cell only:

Theorem 1 (mass-creation purity of Jordan cascades is $k = 2$-only)**.** *For the Jordan block J_k, $k \geq 3$, at any eigenvalue (a complex eigenvalue rescales V by a positive factor, leaving divisor signs unchanged), write $V(t) = f(t^2 N)$ with $f(x) = \sum_m x^m/(m!)^2$ and N the nilpotent. The divisor is $h(N)$ with $h(x) = f(T^2 x)/f(s^2 x)$, $T = s + \tau$, and its second-superdiagonal coefficient is*

$$h_2 = \tfrac{1}{4}\,(T^2 - s^2)(T^2 - 3s^2) \;<\; 0 \quad \textit{for } 1 < T^2/s^2 < 3.$$

In particular every sufficiently short step ($T < \sqrt{3}\,s$) has a negative divisor entry, and the divisor generator satisfies $L_{i,i+2}(t) = -t^3$ exactly in the Jordan basis (a fixed positive multiple $|c_i c_{i+1}|^2$ of it in a general chain, leaving the sign unchanged): no irreducible Jordan cascade of depth ≥ 3—an equal-diagonal, nearest-neighbor bidiagonal chain, the physical object throughout—is negativity-free.

Proof. $|U_{i,i+m}|^2 = t^{2m}/(m!)^2$ for the single block, so $V(t) = f(t^2N)$; functions of N commute, giving h. A general equal-diagonal chain (bonds c_i) is a fixed positive-diagonal similarity $V = D^{-1}f(t^2N)D$ of the block, multiplying each divisor entry by a positive bond factor and leaving every sign invariant. Direct expansion of $f(bx)/f(ax)$ to second order gives $h_2 = b^2/4 - ab + 3a^2/4 = (b-a)(b-3a)/4$ with $a = s^2$, $b = T^2$. For the generator, $L = \dot{V}V^{-1} = 2tN\,(f'/f)(t^2N)$ and $[f'/f]_1 = -\frac{1}{2}$ (or see Theorem 2), so $L_{i,i+2} = 2t \cdot (-\frac{1}{2}t^2) = -t^3$. □

Verified: closed form against direct matrix divisors to 5.9×10^{-13}; the window boundary located at $T = 1.73250$ against $\sqrt{3} = 1.73205$ (grid 5×10^{-4}); the generator entries $2t$ and $-t^3$ to six digits. The conformal-gravity paper's statement is untouched—its cell is $k = 2$, where the single superdiagonal is governed by Proposition 1 and $r = t^2$ is monotone—but stacked sectors do not escape it: the M-quantum sector of the equal-frequency tower carries a *single, irreducible* Jordan block of size $M+1$ (the sector-tower proposition of [4], the constructive form of Smilga's infinite-order exceptional point [11]), so the theorem extends to every sector $M \geq 2$: the sector shift $d\Gamma(N)$ carries strictly positive weights $\sqrt{(\nu+1)(M-\nu)}$ (occupation index $\nu = 0, \ldots, M-1$), so the beable and Jordan bases differ by one fixed, t-independent positive-diagonal similarity at every M [4], under which the entrywise divisor signs are invariant: the tower cannot present a pure mass-creation process—negativity at depth two is a theorem at every sector $M \geq 2$, given the companion's sector-tower structure, for every sufficiently short step.

The full sign structure is Bessel-zero arithmetic:

Theorem 2 (Rayleigh sign law for the cascade generator)**.** *For the Jordan cascade of any depth, at any eigenvalue (a complex eigenvalue shifts only the diagonal of the generator, cf. Theorem 1), the divisor generator's $(m+1)$-th-superdiagonal entries are, in the Jordan basis (a positive bond factor multiplies them in a general chain, leaving signs unchanged),*

$$L_{i,\,i+1+m}(t) \;=\; 2\,(-1)^m\,4^{\,m+1}R_{m+1}\,t^{2m+1}, \qquad R_p = \sum_{n\geq 1} j_{0,n}^{-2p},$$

where $j_{0,n}$ are the positive zeros of J_0 and R_p are the classical Rayleigh sums (exact rationals: $R_1 = \frac{1}{4}$, $R_2 = \frac{1}{32}$, $R_3 = \frac{1}{192}$, $R_4 = \frac{11}{12288}$, …) [10]. The signs strictly alternate at every depth and every $t > 0$.

Proof. $f(x) = \sum_m x^m/(m!)^2 = I_0(2\sqrt{x})$ is entire of order $\frac{1}{2}$ with zeros exactly at $x_n = -j_{0,n}^2/4$, so its Hadamard factorization is $f(x) = \prod_n (1 + 4x/j_{0,n}^2)$ and

$$\frac{f'}{f}(x) = \sum_n \frac{1}{x + j_{0,n}^2/4},$$

a sum of simple poles on the negative axis with unit weights. Expanding each term geometrically, $[f'/f]_m = (-1)^m 4^{m+1} R_{m+1}$, and $L = 2tN\,(f'/f)(t^2N)$ distributes the coefficients along the superdiagonals. Alternation is manifest since $R_p > 0$. □

The exact rational coefficients were confirmed against direct sums over 2×10^5 Bessel zeros, to 5.1×10^{-12} for R_1 (whose slowly convergent sum carries an analytic tail correction) and $\lesssim 3 \times 10^{-15}$ otherwise. At finite steps the same analytic structure governs the windows: the poles of the rescaled divisor $f(\rho y)/f(y)$ sit at the x_n for generic step ratio $\rho = T^2/s^2$ (at the ratios $\rho = (j_{0,k}/j_{0,1})^2$ the numerator vanishes at the dominant pole x_1—precisely the band edges below; cancellations at a

deformation	kind	phase	dynamics / divisor image
Einstein term m_2^2 (Stelle [14])	family	$\varphi = 0$	oscillatory; cost $2/\delta$; negativity returns at $\pi/\lvert\delta\rvert$
loop width (Merlin [12])	family	$\arctan(\gamma/m_2^2)$	growth $\gamma/2\omega$ per pair (time-dilated; see text)
self-interaction at $k = 0$ [6]	corner	$\varphi = \pi$	blowup, rate $(\sqrt{3}/2)\gamma_b$; size-4 block: negativity by Thm 1
non-symplectic lattice / machine noise	generic	random	growing (w.p. 1 for unstructured phase-only noise; fully generic $\to 1$ (rate $1 - O(\sqrt{\epsilon})$, bounded phase density); a structured lattice can split real), rate $\sqrt{\epsilon}$
stacked sectors [11]	—	—	mixed mechanisms (Theorem 1)

Table 1: Deformations of the conformal point, classified by unfolding kind and phase, with the process-level image. The oscillatory set is the punctured real axis ($\Delta \neq 0$) of the family directions (while ω_2^2 stays positive) and, the ray $\varphi = 0$ (the single-entry-model reference direction, resolved to within $O(\epsilon)$) of the generic ones—measure zero in the complex plane either way.

subdominant x_n occur at $(j_{0,k}/j_{0,n})^2$ and shift only subleading terms), the residue at x_1 carries the factor $J_0(j_{0,1}\sqrt{\rho})$, and for large depth m

$$\operatorname{sign} h_m \to (-1)^m \operatorname{sign} J_0\big(j_{0,1}\sqrt{\rho}\big), \qquad h_{m+1}/h_m \to -4/j_{0,1}^2 = -0.691660\ldots$$

(computed -0.6917 by $m \approx 10$ at $\rho = 2, 6$; slower near the band edge at $\rho = 14$). The negative windows of $h_m(\rho)$ converge to *Fourier–Bessel bands* with edges at $(j_{0,n}/j_{0,1})^2 = 5.269,\ 12.949,\ 24.042, \ldots$: the even-$m$ first-window edge marches $3.000 \to 4.778 \to 5.261 \to 5.2689$ ($m = 2, 4, 6, 8$) against 5.269, and the odd-m windows converge onto (5.269, 12.949), reaching (5.269, 12.947) at $m = 11$. (The positivity of h_3 at every step ratio, which a 2×2 intuition would not predict, is exact: $36\, h_3(\rho) = (\rho - 1)(\rho^2 - 8\rho + 19)$ (in the rescaled convention; the dimensional coefficient carries a further positive s^6), and the quadratic factor has negative discriminant; the first odd-m window opens at $m = 5$, inside the band.) These band statements are dominant-pole asymptotics checked numerically; they sit below the strength of Theorems 1 and 2 and are to be read at that strength. The sign map of a deep *Jordan* cascade's divisor thus depends on the depth and the step ratio and on nothing else in the physics.

4 The deformation dictionary

The two layers assemble into a dictionary (Table 1).

Two rows need comment. The *Merlin* row records a result that corrects a natural first guess: one might expect the ghost's loop-induced width γ to impose a detuning floor $|\omega_1 - \omega_2| \gtrsim \Gamma_M$ (the pole width expressed as a frequency, $\Gamma_M = \gamma/m_r$) and so to detune the theory off the exceptional line in the deep UV. It does not. The Donoghue–Menezes pole shift is Lorentz-invariant in q^2 ($M^2 = m_r^2 + \mathrm{i}\gamma$) [12], so at spatial momentum k the frequency-space width $\gamma/2E_k$ time-dilates at exactly the $1/k$ rate of the real detuning $m_2^2/2k$: both components of the complex detuning vanish together, the condition number still diverges (suppressed only by the k-independent factor $[1 + (\gamma/m_2^2)^2]^{-1/2} \approx 1 - 2 \times 10^{-4}$ at $\xi = 1$, $N_{\text{eff}} \approx 20$ in the notation of the revised [4]), and the only change is a fixed $\approx 1^\circ$ tilt of the approach ray into the complex plane. The dressed near-EP is thus of resonance type—one real and one complex frequency at every finite k—and the saturation

a fixed-linewidth intuition would predict ($\mathrm{cond}_{\max} \sim 640\pi/\xi^2 N_{\mathrm{eff}}$) is an artifact of freezing the width in frequency. This paragraph's content appears in the revised companion [4]; we repeat it here because it is the dictionary's only *tilted* family row.

The *stacked sectors* row is Theorem 1 speaking, with its premise supplied by the companion: the M-quantum sector of the equal-frequency Fock tower carries a single irreducible Jordan block of size $M{+}1$ (the sector-tower proposition of [4]; the constructive form of [11]), so—the sector similarity being a fixed, t-independent positive-diagonal one at every M [4], under which entrywise signs are invariant—the tower cannot present a pure mass-creation process; negativity at depth two is a theorem at every sector $M \geq 2$ (for every sufficiently short step), with generator entry a fixed positive multiple of $-t^3$ (exactly $-t^3$ in the Jordan basis). Purity is a property of the minimal cell, and of nothing above it.

5 The photonic readout

Both layers would be measurable with classical light. The program's experimental proposal [5] extracts divisibility observables from the intensity transfer matrices $\hat{\Gamma}(z)$ of lossy waveguide couplers on the Klauck platform [17], and fixes the measurement chain and error model (fabrication disorder, per-launch calibration, shot noise, dark floor) that we adopt unchanged here; of the two devices below, D5 (the unidirectional cascade cell) is new, while D4 adds a new guide-basis negativity readout to the proposal's existing EP coupler; their error budget is stated in full in this section and in the public script.

D4: the dimer at its own EP ($\gamma = 2\kappa$). A reciprocal coupler's Jordan vector at the EP is the superposition $(1, -\mathrm{i})/\sqrt{2}$, not a guide, so the guide-basis process is *not* a cascade and fails by negativity, with onset $\to 1/(\kappa\sqrt{2})$ (a convention-free negativity onset, not the mass-creation t_c of the conventions paragraph), the continuous EP limit of the broken-phase onset formula (at $\kappa = 0.30\,\mathrm{cm}^{-1}$: simulated onset 2.15 cm at the 0.25 cm divisor step, against the formula's 2.36 cm—the ∼0.2 cm shortfall tracking the finite divisor-step discretization, of which the half-step $\tau/2$ anchor used for D5 accounts for about half).

D5: the cascade cell in glass. Realizing the PU cell in the beable (guide) basis requires *unidirectional* coupling: coherent coupling J at synthetic phase $\pi/2$ plus dissipative coupling Γ_d through a common lossy bus gives $H_{12} = \mathrm{i}(J - \Gamma_d)$, $H_{21} = -\mathrm{i}(J + \Gamma_d)$, one-way at $J = \Gamma_d$ (a Hatano–Nelson asymmetry [18]). The device shows the mechanism fingerprint on two-length divisor data: the det-based onset estimator of the original proposal is *provably blind* (a triangular divisor's determinant is a product of positive diagonal ratios) while the column sum crosses one at $z_c = 0.91\,\mathrm{cm}$ ($J = \Gamma_d = 0.25\,\mathrm{cm}^{-1}$, divisor step 0.25 cm; the onset is protocol-dependent, $0.486/2J = 0.97\,\mathrm{cm}$ in the zero-step limit, and convention-relative in the sense of Sec. 3: it is the $\sigma_{\max}$-normalized observable). The normalization is itself measurable here: for the triangular device $U^\dagger U$ has diagonal equal to the measured column sums of $\hat{\Gamma}$ and off-diagonal modulus $\sqrt{\hat{\Gamma}_{21}\hat{\Gamma}_{22}}$, so $\sigma^2_{\max}(z)$ is the largest eigenvalue of a matrix assembled from the measured intensities alone—no phase-resolved characterization is needed. This reconstruction is exact at triangularity (relative error 6×10^{-16}) and, under asymmetry imperfection, accurate to the first-order data-driven Weyl estimate $\sqrt{\hat{\Gamma}_{11}\hat{\Gamma}_{12}}$ set by the measured residual entry itself (relative 5×10^{-3} at $\eta = 3\%$ of J; script block [1c]). Det-silent but column-sum-loud identifies the cascade device of this pair; det-crossing with negativity, the dimer. Detuned variants ($\delta = 0.5$ and $0.8\,\mathrm{cm}^{-1}$) read the direction: negativity first returns near $z^* = \pi/\delta$ (simulated: 6.16 and 3.81 cm at the 0.25 cm divisor step, against $\pi/\delta = 6.28$ and 3.93 cm—for z^* the offset is exactly half the pair spacing: $\pi/\delta - \tau/2 = 6.158$ and 3.802 cm). Simulated at 10^4 detected photons per launch with the proposal's error model (400

realizations), using closed-form, calibration-free estimators for the balanced device (Γ_d from the log-slope of the dark-subtracted $\hat{\Gamma}_{11}$ ladder; $J + \Gamma_d$ from $\sqrt{\hat{\Gamma}_{21}/\hat{\Gamma}_{11}}/z$, in which the per-launch calibration cancels exactly): J recovered to $+0.5\% \pm 1.7\%$ with recovery correlation 0.74, printed as a control—an earlier least-squares model fit returned only its initialization on the balanced device, so its apparent precisions were echoes of the injected disorder priors, and the script now prints that non-recovery ($\mathrm{corr}(\hat{\eta}, \eta) = 0.04$) as a permanent control; $\hat{z}_c$ to $\pm 0.040\,\mathrm{cm}$ at the fixed 0.25 cm protocol (the protocol convention itself shifts the onset by 0.06 cm, as stated above, and is part of the quoted observable); and z^* to $\pm 0.22\,\mathrm{cm}$ about the finite-step offset (simulated center $-0.10\,\mathrm{cm}$ vs the $-\tau/2 = -0.125\,\mathrm{cm}$ anchor; the detuned-device fit that delivers it is itself controlled, $\mathrm{corr}(\hat{\delta}, \delta) = 0.86$, printed). The asymmetry imperfection η (convention: back-coupling $H_{12} = \mathrm{i}\eta$ at fixed $H_{21} = -\mathrm{i}(J + \Gamma_d) = -0.5\mathrm{i}\,\mathrm{cm}^{-1}$) is *not* resolved by the intensity fit at the 3% design level—its $\hat{\Gamma}_{12}$ signal peaks at the dark floor at these photon numbers—so it is certified rather than fitted, and the certification is necessarily *two-channel*, because the negativity channel reads only one sign of η: negativity-cleanliness of a 6 cm window bounds the real-split side η/J ($\eta > 0$) below the ≈13% boundary (computed: $z^* = 6.0\,\mathrm{cm}$ at 13%, first in-window negativity at 14% on the 1% grid near the boundary, boundary between 13 and 14%; the z^* estimator scans pair base points at 0.01 cm with 0.25 cm spacing, the window-negativity scan at 0.1 cm—window-aligned conventions shift these by up to one grid step), and larger positive imperfections are measured directly by the re-entry length (computed: $z^* = 6.9\,\mathrm{cm}$ at $\eta/J = +10\%$, against 12.7 cm—off-chip—at +3%; the imperfect $H_{12} \neq 0$ device is non-triangular, so the triangular bond rule $z^* = \pi/|\delta|$ no longer governs and onset falls below it). The imaginary-split side ($\eta < 0$) is negativity-*silent* at any magnitude (the imaginary split keeps the relevant two-time determinant ≥ 1 throughout, so the sign flip never triggers; corroborated computationally, no in-window negativity at $\eta/J = -14\%$ through -100%), so window cleanliness alone cannot bound it; that side is bounded instead by the Weyl residual of the intensity reconstruction: the error in $\sigma^2_{\max}$ obeys $\lesssim \sqrt{\hat{\Gamma}_{11}\hat{\Gamma}_{12}}$ (script block [1c]; first order in the residual, the $O(\hat{\Gamma}_{12})$ correction absorbed by the margin below), which is sign-blind in η—it reads the measured back-coupling intensity $\hat{\Gamma}_{12}$ directly—and bounds the triangularity systematic from the same data. The proposal's 3% per-launch calibration level is thus met with a factor-four margin (heuristically, to first order in the residual) against the 13% boundary on the real-split side, and by the sign-blind intensity residual on the silent side. The imperfection's *sign* selects the unfolding direction ($\eta > 0$: real split, negativity re-enters at finite z; $\eta < 0$: purely imaginary split, clean past 30 cm): even the device's error budget is direction physics.

For feasibility we note that at-EP process physics is experimentally live (an observation of quantum exceptional points of an extended Liouvillian on a superconducting circuit appeared in 2025 [20]—non-Markovian EPs, fittingly for this program), that Klauck-type arrays reach the required loss rates, and that the required asymmetric (Hatano–Nelson) coupling has been realized in photonics: Weidemann et al. [19] implemented tunable anisotropic coupling in a non-Hermitian photonic mesh lattice—the non-Hermitian skin effect experiment—in the same experimental school as the Klauck platform. Transferring that coupling asymmetry from the temporal mesh to a laser-written waveguide pair via a common lossy bus is a design exercise the proposal's error budget estimates—a single coupling transfer between demonstrated platforms, itself not yet tested end-to-end, rather than a new capability.

6 Relation to prior work

The Puiseux mathematics of EP splitting is classical [8] and its physics literature is active—a direction-dependent Puiseux classification via Hessenberg forms of the perturbation, with sensing applications [24] (the closest prior art to Sec. 2's classification, carrying no divisibility or process content), formalism at exceptional points [21], algebraic characterizations of non-Hermitian degeneracies [22], and, closest in phrase, "unitary paths of unfolding" through a third-order EP in a solvable model [23], which exhibits particular reality-preserving corridors where we classify the finite-frequency family's deformation directions and attach a stochastic-mechanism image. The closest EP–divisibility contact in the literature is the non-Markovian exceptional-point line [20, 16, 27], where EPs emerge in open-system spectra tied to divisibility transitions (most directly in [27], whose channel-level EP explicitly violates CP-divisibility); that line classifies neither unfolding directions nor divisor failure mechanisms. On the gravity side, Kumar and Marto [25] deliberately trade the Stelle ghost for an inverted-harmonic-oscillator instability and quantize on that side; in the present language theirs is a committed choice of the growing side of the same boundary, and the two sides are related by the unfolding direction of one Jordan block rather than by separate constructions. In the divisibility literature, the geometric characterization of stochastic divisors by Pimenta [26] is trace-preserving throughout—unit column sums are structural there—so the column-sum-crossing mechanism central to this paper and to [4] lies outside that framework by construction; our sub-stochastic setting is precisely what makes conditional mass creation available. The resonant-PU algebraic thread of Fring, Marquette and Taira [28] is engaged in [4]; the present paper adds the divisor-level image their algebra does not address. Ilhan and Kovner's two papers [6, 7] supply the corner instability and the Jordan-block statement respectively; our use of them is scoped in Sec. 2. To our knowledge no prior work attaches a stochastic divisibility-mechanism image (a divisor sign-and-column-sum readout) to the directional unfolding of an exceptional point—the directional Puiseux classification itself has independent precedent [24], and the divisor image is what is new here—and no photonic proposal measures divisor column sums; we date these claims to the program's public notebook in the program's hub repository.

7 Discussion

The phase-boundary reading changes what kind of object conformal gravity is. (We use "phase boundary" in the exceptional-point sense—dynamical regimes separated by a spectral degeneracy—not in the Landau sense of an order parameter; the analogy is structural, and we flag it as such.) On this map the pure theory occupies a measure-zero set: every physical deformation, every regulator, and every numerical representation sits on the growing, oscillatory, or (for net-damping representations) decaying side, and the side is decided by a phase. The theory's notorious fragility at the equal-frequency point—polynomial growth flipping to exponential under machine-epsilon perturbations, recorded in [4]—is not a numerical nuisance but the boundary itself, and the classification converts anecdotes into a dictionary (Table 1) with a per-row observable.

The divisor theorems give the boundary a laboratory-facing signature and settle an internal question. Mass-creation-pure indivisibility—the mechanism that the conformal-gravity paper found at the conformal cell—is now known to be a property of the minimal Jordan cell exclusively: depth-three *Jordan* cascades are negativity-bearing as a matter of Bessel-zero arithmetic, with exactly alternating generator signs (a reducible three-level cascade, e.g. a $k = 2$ cell plus a spectator level, can of course remain pure—the theorem is about the irreducible block). The universality is striking: the sign map of a deep Jordan cascade's divisor knows the depth and the step ratio,

and nothing else about the physics. We do not currently know a physical principle that singles out the Rayleigh sums here beyond the calculation itself; the appearance of Fourier–Bessel band edges $(j_{0,n}/j_{0,1})^2$ in a divisibility question seems to us genuinely new, and possibly a shadow of something better.

In the program's arc, this paper is the second of the pair the threads audit projected: the conformal-gravity paper proved what fails *at* the conformal point; this paper proves what the neighborhood looks like and how to measure which neighbor nature chose—if a fourth-order sector is physically realized at all (the depth-two negativity is a theorem for the formal Jordan block regardless), which remains the program's standing conditional. The natural continuations are the finite-time and dressed-propagator questions (the sheet classification of ghosts versus unstable particles admits a process-level image we leave to later work), and the boundary conditions of the de Sitter completion, where the same Jordan degeneration prices Maldacena's ghost-removal [15]; both are planned as later installments of the program.

Code availability. All numerical claims are reproducible from the public scripts in the program repository—except the Merlin-row loop numbers of Sec. 4, which are analytic computations whose arithmetic is stated in the revised companion [4]: `scripts/p6_unfolding_sweep.py` (Sec. 2), `scripts/p6_divisor_gate.py`, `scripts/p6_divisor_theorem.py`, `scripts/p6_divisor_dxd.py`, `scripts/p6_cm_signs.py`, `scripts/p6_m2_channel.py` (Sec. 3), and `scripts/p6_d4_device.py` (Sec. 5), each printing its controls inline.

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